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SYSTEMS, METHODS, AND DEVICES FOR AUTOMOBILE ENVIRONMENT MONITORING AND SIGNAL STEERING

Abstract

Systems, methods, and devices provide monitoring of entities in automobile environments. Methods include determining, using a processing device, a first plurality of phase modulation parameters for a speaker array based on a relationship between speakers included in the speaker array and a designated target location. Methods further include generating, using the processing device, a plurality of audio signals for the speakers included in the speaker array, each of the plurality of audio signals including a phase adjustment determined based on the first plurality of phase modulation parameters, and transmitting the plurality of audio signals via the speaker array to the designated target location.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to automobile environments, and more specifically, to enhancement of monitoring of entities and signal steering in such automobile environments.

BACKGROUND

[0002] Infotainment systems and wireless devices may be implemented in a variety of contexts, such as automobiles and other vehicles. Such systems and devices may be configured to support wireless communications in accordance with wireless communication protocols. Moreover, they may also include various other sensing modalities to obtain information about an operational environment, such as whether or not a person is present within the operational environment. However, conventional techniques for transmitting and receiving audio signals as well as performing radar detections remain limited because they are not able to efficiently perform such operations in various operational conditions such as, for example, ambient noise is high.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** illustrates diagram of a system for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0004] FIG. **2** illustrates diagram of a device for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0005] FIG. **3** illustrates diagram of another system for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0006] FIG. **4** illustrates diagram of an additional device for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0007] FIG. **5** illustrates diagram of another system for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0008] FIG. **6** illustrates diagram of an additional system for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0009] FIG. **7** illustrates diagram of a method for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0010] FIG. **8** illustrates diagram of another method for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0011] FIG. **9** illustrates diagram of an additional method for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0012] FIG. **10** illustrates diagram of another method for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

[0013] FIG. **11** illustrates diagram of a method for automobile environment monitoring and signal steering, configured in accordance with some embodiments.

DETAILED DESCRIPTION

[0014] In the following description, numerous specific details are set forth in order to provide a thorough understanding of the presented concepts. The presented concepts may be practiced without some or all of these specific details. In other instances, well known process operations have not been described in detail so as not to unnecessarily obscure the described concepts. While some concepts will be described in conjunction with the specific examples, it will be understood that these examples are not intended to be limiting.

[0015] Embodiments disclosed herein provide signal steering for signals transmitted and received within an operational environment, such as the interior of an automobile. As will be discussed in greater detail below, arrays of speakers and microphones may be configured to provide beam forming and signal steering for audio signals used for transmitting audio and receiving audio within the operational environment. As will also be discussed in greater detail below, such improved techniques for delivery and reception of audio signals may also be leveraged to improve performance of radar capabilities of components included within the operational environment. For example, such arrays of speakers and microphones may be used to support radar operations used for in-cabin monitoring and seat-occupancy-detection.

[0016] FIG. **1** illustrates diagram of a system for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, an automobile environment, such as automobile environment **100**, may include various components configured to monitor a status of one or more locations within automobile environment **100**. As will be discussed in greater detail below, automobile environment **100** includes components configured to utilize such status information to enhance transmission to and reception from such locations. Moreover, such directional signal steering allows automobile environment **100** to make presence determinations about specific locations within automobile environment **100** and target delivery of an audio signal to such specific locations as well as receive an audio signal from such specific locations.

[0017] In various embodiments, automobile environment **100** is a vehicle, such as an automobile, that includes multiple seating locations, such as seat location **114**, seat location **116**, seat location **118**, seat location **120**, and seat location **122**. Automobile environment **100** additionally includes components configured to monitor the different seat locations, and determine whether or not an entity, such as a user, is present at the seat locations. In various embodiments, the monitoring may be performed via one or more radar operations in which sound is transmitted and received, and a presence of an entity is determined based on a reflected signal.

[0018] Accordingly, automobile environment **100** includes radar module **104** which is configured to transmit and receive a signal that may be generated based on a reflection of the transmitted signal. In various embodiments, radar module **104** includes a transducer, such as a microphone, as well as associated processing hardware, such as an amplifier, configured to receive and amplify the received signal. For example, radar module **104** may be configured to receive a signal at 60 GHz, and generate an output based on such a received signal. In some embodiments, radar module **104** is included in an in-cabin monitoring system provide by Infineon®. In some embodiments, radar module **104** additionally includes one or more processors and a memory that may be configured to perform radar detection operations. Accordingly, radar module **104** may be configured to determine if an entity is present based on one or more temporal, frequency, and/or phase computations performed in accordance with any suitable radar computation. For example, radar module **104** may be a Doppler radar that transmits periodic pulses, and performs ranging computations based on a reflected signal. More specifically, a temporal offset value determined based on the timing of the reception of the reflect signal may be compared against a threshold value to determine if an entity is present at a designated location. Moreover, radar module **104** may be configured to generate an output representing a result of such a determination.

[0019] In various embodiments, radar module **104** is configured to use a Frequency Modulated Continuous Wave (FMCW) transmitted over a range (R) to one or more targets within a Field of View (FoV) can be calculated using equation 1 shown below:

[00001] $R = \frac{T_C \times \ f \times C}{2 \times \mathrm{BW}} \quad (1)$

[0020] In equation 1, T.sub.C is a chirp period of the radar signal, Δf is the receive beat frequency, C is the speed of light, and BW is the occupied bandwidth of the radar signal. In various embodiments, such parameters may be determined by an entity, such as a manufacturer in accordance with operational parameters determined during a design process. In some embodiments, to determine angular offsets of radar targets within the FoV, the transceiver may use two or more receiver antennas. In some embodiments, virtual receiver antennas (n.sub.RX) may be determined based on the number of Tx antennas multiplied by the number of receiver antennas. In some embodiments, the angular resolution (ΔA) is improved with a larger number of n.sub.RX, as shown in equation 2 below, where λ=the wavelength of the radar signal:

[00002] $$A = \frac{}{d \times n_{\mathrm{RX}} \cos} \times \frac{180}{} \quad (2)$$

[0021] Thus, according to some embodiments, to detect occupants within the cabin of a vehicle, FMCW modulation is used, and a transceiver included in radar module **104** may use two transmit channels and four receiver channels, or eight virtual receiver antennas. Therefore, when the transceiver is operating at 60 GHz the ΔA is then 14.3 degrees. In various embodiments, such operational parameters are capable of detecting the position of five adults within the cabin of a five-seater vehicle with an accuracy of 90% or better.

[0022] Automobile environment **100** additionally includes processing device **102** which includes one or more processors and a memory that may be configured to perform processing operations for automobile environment **100**, such as management and streaming of data. In various embodiments, processing device **102** is a head unit, as may be included in an infotainment system of an automobile. Accordingly, processing device **102** may include additional hardware and software for wireless communication, such as one or more wireless transceivers and associated processing logic. Moreover processing device **102** may be configured to generate an audio signal that is sent to speaker **106**, speaker **108**, speaker **110**, and speaker **112**. In some embodiments, processing device **102** is configured to execute an application used to stream audio data, and such audio data may be converted to an audio signal provided to the speakers. Moreover, processing device **102** may also be configured to generate a signal sent via the speakers during a radar detection mode that is used to augment radar detection operations of radar module **104**.

[0023] Automobile environment **100** one or more speakers, such as speaker **106**, speaker **108**, speaker **110**, and speaker **112**. As will be discussed in greater detail below, each speaker may be a speaker array configured to generate an audio signal targeted at a seat location in a configurable manner. For example, each speaker may perform one or more beam forming operations to guide an audio signal to the targeted location. As will be discussed in greater detail below with reference to FIG. **2** and FIG. **3**, speaker **106**, speaker **108**, speaker **110**, and speaker **112** may collectively target a single seat location such that a received signal and/or radar determination is made specific to that seat location.

[0024] In various embodiments, the beam forming and targeting may be configured to cycle through target seat locations in a designated order. Moreover, such an order may be configurable by an entity, such as a manufacturer or a user/occupant of the vehicle. For example, a user may specify a priority list of seat locations that is used to determine an order in which target seat locations are scanned. In one example, if an occupant of automobile environment **100** is expecting a phone call, the occupant may identify himself or herself as having the highest priority in beam forming operations. Such a higher priority may correspond to increased priority in a scanning order, as well as increased time spent targeting that seat location. Such a priority determination may also selectively omit other seat locations. In the example of an expected phone call, rear seat locations may be omitted as they may be occupied by children.

[0025] Moreover, such configuration of beam forming operations in the context of seat locations may be configured dynamically and based on events within automobile environment **100**. For example, reception of a phone call, as may be detected by processing device **102**, may trigger implementation of a designated priority list, such as omission of rear occupants and a highest priority for the driver. Similarly, types data values and audio streams detected by processing device **102** may also be used to implement different priority lists. For example, if a movie or particular type of audio file is detected as being played at processing device **102**, the rear occupants may be given highest priority and the front driver and passenger may be omitted. In this way, beam forming and targeting of audio signals may be configured based on designated parameters set by a user or manufacturer, or configured dynamically based on events within automobile environment **100**.

[0026] FIG. **2** illustrates diagram of a system for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, speakers used for audio signal transmission and radar detection operations may be configured to generate directional audio signals configured to target specific locations. As will be discussed in greater detail below, such speakers may include speaker arrays that are configured to generate directional audio signals used to target locations, and such targeting may be determined based on one or more radar determinations.

[0027] Thus, according to some embodiments, automobile environment **200** includes processing device **202**. As similarly discussed above, processing device **202** may be included in a head unit of an automobile and may include one or more processors configured to make presence determinations identifying the presence or absence of an entity at a target location. In various embodiments, automobile environment **200** additionally includes audio source **204** which may be configured to generate an audio signal transmitted by one or more speakers. As similarly discussed above, audio data may be generated by an application executed on a device, such as audio source **204**, and such audio data may be transmitted via speaker array **207**. In some embodiments, audio source **204** may also include a signal generator and associated processing logic configured to generate an audio signal used for radar detection operations. The audio signal may be an arbitrary waveform, or a waveform generated in accordance with a radar standard and configured based on capabilities of the speakers. It will be appreciated that audio source **204** may be included within processing device **202** or implemented separately.

[0028] Automobile environment **200** additionally includes phase modulator **206** which is configured to generate phase modulation parameters used to implement directional targeting of an audio output of a speaker array, such as speaker array **207**. As will be discussed in greater detail below, such phase modulation parameters may be phase offset values computed for each speaker.

[0029] In various embodiments, speaker locations and seat locations remain fixed and relatively constant within a vehicle other than relatively small lateral movements associated with the front driver and passenger seats due to vehicle motion. Accordingly, an angle from any speaker to any seat location is known based on the geometry of the vehicles cabin design. As similarly discussed above, radar techniques can determine if a given seat location is occupied by a person, and the location of the occupied seat can then be sent from a component, such as a radar module, to speaker locations. As will be discussed in greater detail below, each speaker may have a look-up-table that may be stored in memory or in an EEPROM of an associated microcontroller unit (MCU) that may be included in processing device **202**. In various embodiments, the angle from the speaker to the occupied seat location can then be determined based on the data values stored in the look-up-table. In some embodiments, the phase shift (@) for each speaker is then calculated using equation 3 shown below. In equation 3, d is the distance between each speaker in the speaker array, AoA is the angle of arrival for the audio signal to the target, and λ is wavelength of the audio signal.

[00003] $$= \frac{2 \times \quad \times d \times \sin AoA\left(\frac{}{180}\right)}{} \quad (3)$$

[0030] More specifically, speaker array **207** may include multiple different speakers configured to generate an audio output based on a received input signal. For example, speaker array **207** may include speaker **210**, speaker **214**, speaker **218**, speaker **222**, which may each have an associated amplifier, such as amplifier **208**, amplifier **212**, amplifier **216**, and amplifier **220**. As will be discussed in greater detail below, the phase modulation parameters may configure a directionality of an output of speaker array **207**. In this way, such phase modulation parameters may be modified to target different locations within automobile environment **200**.

[0031] FIG. **3** illustrates diagram of another system for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, automobile environment **300** may include radar module **304** and processing device **302**. Automobile environment **300** may also include seat location **314**, seat location **316**, seat location **318**, seat location **320**, and seat location **322**. Automobile environment **300** may further include various speakers, such as speaker **306**, speaker **308**, speaker **310**, and speaker **312**. As also discussed above, each speaker may be a speaker array configured to directionally transmit an audio signal based, at least in part, on phase modulation parameters.

[0032] As shown in FIG. **3**, audio signals from the speakers may be collectively targeted at a particular location. In this example, speaker **306**, speaker **308**, speaker **310**, and speaker **312** are all targeted at seat location **322**. In one example, radar module **304** may have performed one or more radar operations and determined an entity is present at seat location **322**. Radar module **304** may have provided this determination to processing device **302**, and processing device **302** may have determined phase modulation parameters for each of the speakers based on this determination. Processing device **302** may then configure a phase modulator for each speaker, which may be each be a speaker array. Processing device **302** may also provide an audio signal for transmission to the speakers. For example, audio signal **324** may be transmitted from speaker **306**, audio signal **326** may be transmitted from speaker **308**, audio signal **328** may be transmitted from speaker **310**, and audio signal **330** may be transmitted from speaker **312**.

[0033] In some embodiments, the speakers may participate in the radar determinations. For example, a first target location may be selected, such as seat location **322**. Moreover, audio signal **324** may be transmitted from speaker **306**, audio signal **326** may be transmitted from speaker **308**, audio signal **328** may be transmitted from speaker **310**, and audio signal **330** may be transmitted from speaker **312**. A reflected signal may be received at radar module **304** and used to determine if an entity is present. In this example, during radar detection operations, the speakers may cycle among seat locations based on designated phase modulation parameters such that each seat location is measured periodically.

[0034] FIG. **4** illustrates diagram of an additional device for automobile environment monitoring and signal steering, configured in accordance with some embodiments. In various embodiments, an automobile environment, such as automobile environment **400**, may be configured to include a microphone array, such as microphone array **408**, that is configured to receive audio signals directionally. In this way, one or more dedicator microphone arrays may be included in an automobile environment to improve reception of audio signals that may originate from, for example, a passenger seated at a particular location. In some embodiments, such directional audio signal detection may also be used for radar operations.

[0035] In various embodiments, microphone array **408** may include multiple microphones, such as microphone **410**, microphone **414**, microphone **418**, and microphone **424** which may each have an associated amplifier, such a amplifier **409**, amplifier **412**, amplifier **416**, and amplifier **420**. Accordingly, each microphone may be configured to receive an audio signal that originates within automobile environment **400**, and generate an output based on the received signal.

[0036] In various embodiments, automobile environment **400** additionally includes processing device **402** which is configured to receive audio signals from microphone array **408** and is also configured to retrieve data values from memory **406**. In some embodiments, a signal generated by the radar module that is sent to the speakers may also be sent to the microphones. In some embodiments, a look-up-table is stored in memory **406**, which may be an EEPROM, that determines the angle of each microphone to the location of the target detected by the radar module. As the distance between each microphone in the microphone array is known based on a geometry of the design and/or package of microphone array **408** and the vehicle, the angle of arrive (AoA) of the audio from the occupant in the known seat location is then determined. The phase (@) is then calculated using equation 4 below, where d is the distance between each microphone, and **2** is the wavelength of the audio signal.

[00004] $= \frac{2 \times \quad \times d \times \sin}{\quad}$ (4)

[0037] FIG. **5** illustrates diagram of another system for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, automobile environment **500** may include radar module **504** and processing device **502**. Automobile environment **500** may also include seat location **514**, seat location **516**, seat location **518**, seat location **520**, and seat location **522**. Automobile environment **500** may further include various speakers, such as speaker **506**, speaker **508**, speaker **510**, and speaker **512**.

[0038] In various embodiments, automobile environment **500** additionally includes microphone array **524**. As shown in FIG. **5**, microphone array **524** includes an array of microphones, such as microphone **526**, microphone **528**, microphone **530**, and microphone **532**. In various embodiments, microphone array **524** is configured to receive audio signals from a particular location within automobile environment **500**, such as seat location **522**. As also shown in FIG. **5**, an angle of arrival and a time of arrival at each of the microphones may be different based on differences in their physical locations. Accordingly, phase modulation parameters may be used to interpret such phase and time differences to generate a received audio input from the target location. In various embodiments, the audio signal generated at seat location **522** may be an audio command generated by a user. In some embodiments, the audio signal may be a reflected signal generated during radar operations.

[0039] FIG. **6** illustrates diagram of an additional system for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, automobile environment **600** may include radar module **604** and processing device **602**. Automobile environment **600** may also include seat location **614**, seat location **616**, seat location **618**, seat location **620**, and seat location **622**. Automobile environment **600** may further include various speakers, such as speaker **606**, speaker **608**, speaker **610**, and speaker **612**. As also discussed above, each speaker may be a speaker array configured to directionally transmit an audio signal based, at least in part, on phase modulation parameters. In various embodiments, automobile environment **600** additionally includes microphone array **632**. As shown in FIG. **6**, microphone array **632** includes an array of microphones, such as microphone **634**, microphone **636**, microphone **638**, and microphone **640**. Thus, according to various embodiments disclosed herein, audio may be transmitted directionally within automobile environment **600**, as shown by audio signal **624**, audio signal **626**, audio signal **628**, and audio signal **630**. Moreover, audio may be received directionally as well via microphone array **632**. In this way, audio may be transmitted and received directionally within automobile environment **600** to improve audio delivery and reception with respect to a particular location as well as radar operations specific to a particular location. It will be appreciated that while FIG. **6** shows both the speakers and microphone array **632** targeting the same location, they may also target different locations independently.

[0040] FIG. **7** illustrates diagram of a method for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, speakers may be configured to generate directional audio signals configured to target specific locations within an automobile environment. Accordingly, a method, such as method **700**, may be performed to generate directional audio signals used to target such locations to improve the quality of audio signals received at a target location.

[0041] Method **700** may perform operation **702** during which location information may be determined based on one or more radar operations. As similarly discussed above, a radar module may be configured to perform radar operations to identify the presence of an entity within an automobile environment. Accordingly, during operation **702**, radar operations may be performed, and a radar module may identify the presence of an entity at a designated location. In one example, the radar module may be configured to monitor a set of known locations, such as seating locations, and may periodically scan each of the known locations. In response to identifying the presence of an entity, the known location being scanned may be returned as location information which may be represented by a unique identifier.

[0042] Method **700** may perform operation **704** during which a plurality of phase modulation parameters may be generated based, at least in part, on the location information. Accordingly, as similarly discussed above, a look-up-table may be used to retrieve phase modulation parameters based on the location information. Thus, according to some embodiments, the look-up-table may be configured to map the location identifier to a plurality of sets of phase modulation parameters. For example, a set of phase modulation parameters may be retrieved for each speaker within the automobile environment. Such phase parameters included in the look-up-table may have been determined by an entity, such as a manufacturer, during a design and manufacturing process.

[0043] Method **700** may perform operation **706** during which a plurality of audio signals may be generated based, at least in part, on the phase modulation parameters. Accordingly, audio signals may be generated for each speaker within each speaker array. The audio signal may be audio data, such as music, and each audio signal may be phase modulated based on the phase modulation parameters.

[0044] Method **700** may perform operation **708** during which the plurality of audio signals may be transmitted via a plurality of speakers. Accordingly, the audio signals may be transmitted from the speaker arrays, and directional beam forming may be provided via the phase modulation introduced by the phase modulation parameters. In this way, delivery of the audio signal to the target location may be improved without the use of additional global adjustments, such as global volume adjustments.

[0045] FIG. **8** illustrates diagram of another method for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, speakers may be configured to generate directional audio signals configured to target specific locations within an automobile environment. Moreover, a method, such as method **800**, may be performed to generate directional audio signals that may be used to improve radar operations within an automobile environment.

[0046] Method **800** may perform operation **802** during which a plurality of phase parameters may be determined for at least one speaker array. As similarly discussed above, such phase parameters may be determined based on a look-up-table configured for the at least one speaker array, and the phase parameters may be determined by an entity, such as a manufacturer, during a design and manufacturing process. In various embodiments, an automobile environment may include multiple speaker arrays, such as four speaker arrays. It will be appreciated that each speaker array may have a look-up-table configured specifically for that speaker array based on its position relative to locations within the automobile environment, such as seat locations.

[0047] Accordingly, during operation **802** a location may be identified, and phase parameters may be looked up based on the initial location that was identified. As similarly discussed above, a set of designated locations may be mapped to appropriate phase parameters within the look-up-table, and the phase parameters may be identified based on the mapping. In some embodiments, the location may be identified based on an input received from the automobile environment. For example, such an input may be received from a sensor, such as a seatbelt sensor or a pressure sensor, included at a particular location, such as a seat location. In another example, the location may be identified based on a progression through a designated set of locations. For example, seat locations may be stored as a designated set of locations, and method **800** may cycle through the set of locations to periodically scan each one. Such cycling may be controlled by a state machine or any other suitable means.

[0048] Method **800** may perform operation **804** during which an audio signal may be generated based, at least in part, on the plurality of phase parameters. Accordingly, a waveform may be generated for radar operations, and may be modulated based on the phase parameters. For example, each speaker within the speaker array may have its own audio signal configured based on a phase parameter specific to that speaker. More specifically, the phase parameter may identify a phase shift or phase delay, and the audio signal for each speaker within the speaker array may be modified to implement the phase parameter for that speaker as determined based on the look-up-table.

[0049] Method **800** may perform operation **806** during which the audio signal may be transmitted via the speaker array within an automobile environment. Accordingly, the audio signal may be amplified and output via the speakers included in the speaker array. As discussed above, the phase parameters applied to the audio signals may provide beam forming and directionality of the audio output that targets a designated location that corresponds to the initially identified location mentioned above with reference to operation **802**.

[0050] Method **800** may perform operation **808** during which an audio signal may be received based on one or more interactions between the transmitted audio signal and the automobile environment. Accordingly, the transmitted signal may reflect off of objects included in the automobile environment. For example, the transmitted audio signal may reflect off of an entity, such as a person, seated at the location being measured. Accordingly, the transmitted signal may reflect off of the person and be reflected back to one or more microphones included in a radar module that receives the reflected signal as a received audio signal.

[0051] Method **800** may perform operation **810** during which one or more environmental parameters may be determined based on the received audio signal. In various embodiments, the environmental parameters may be generated based on radar computations, and may identify whether or not an entity is present at the location that was scanned. More specifically, the environmental parameters may be configured to represent a positive or negative determination of a presence of an entity based on radar operations, and may be stored in memory.

[0052] Method **800** may perform operation **812** during which one or more operations may be performed based on the one or more environmental parameters. In various embodiments, the operations may be performed responsive to the detection of an entity. For example, in response to detecting an entity, such as a person, at the location being measured, a message may be generated and transmitted wirelessly to a wireless device, such as a smart phone. In this example, the person may be a child or pet, and the environmental parameters may identify child occupancy. In response to detecting the child, operations such as generation of a notification message, lowering of automobile windows, and/or unlocking of doors may be performed.

[0053] FIG. **9** illustrates diagram of an additional method for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, microphones may be configured to directionally receive audio signals generated at specific locations within an automobile environment. Accordingly, a method, such as method **900**, may be performed to directionally receive audio signals from such locations to improve the quality of audio signals received from such locations and improve tolerance of ambient noise as may occur in an automobile in motion.

[0054] Method **900** may perform operation **902** during which location information may be determined based on one or more radar operations. As similarly discussed above, a radar module may be configured to perform radar operations to identify the presence of an entity within an automobile environment. Accordingly, during operation **902**, radar operations may be performed, and a radar module may identify the presence of an entity at a designated location. In one example, the radar module may be configured to periodically scan each of the known locations. In response to identifying the presence of an entity, the known location being scanned may be returned as location information which may be represented by a unique identifier.

[0055] Method **900** may perform operation **904** during which a plurality of phase modulation parameters may be generated based, at least in part, on the location information. Accordingly, as similarly discussed above, a look-up-table may be used to retrieve phase modulation parameters based on the location information. Thus, according to some embodiments, the look-up-table may be configured to map the location identifier to a plurality of sets of phase modulation parameters. For example, a phase modulation parameter may be retrieved for each microphone within a microphone array. Such phase parameters included in the look-up-table may have been determined by an entity, such as a manufacturer, during a design and manufacturing process.

[0056] Method **900** may perform operation **906** during which an audio signal may be received at a plurality of microphones and based on the plurality of phase modulation parameters. Accordingly, an audio signal may be generated at the location being observed by, for example, a seated passenger speaking, and the audio signal may be received at the microphones included in the microphone array. The audio signal generated by the microphones may be phase modulated based on the phase modulation parameters to provide beam forming and directionality for listening capabilities of the microphone array.

[0057] FIG. **10** illustrates diagram of another method for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, microphones may be configured to directionally receive audio signals from specific locations within an automobile environment. Moreover, a method, such as method **1000**, may be performed to directionally receive audio signals that may be used to improve radar operations within an automobile environment.

[0058] Method **1000** may perform operation **1002** during which a plurality of phase parameters may be determined for at least one microphone array. As similarly discussed above, such phase parameters may be determined based on a look-up-table configured for the at least one microphone array, and the phase parameters may be determined by an entity, such as a manufacturer, during a design and manufacturing process. It will be appreciated that a microphone array may have a look-up-table configured specifically for that microphone array based on its position relative to locations within the automobile environment, such as seat locations.

[0059] Method **1000** may perform operation **1004** during which an audio signal may be generated within an automobile environment. As similarly discussed above, a waveform may be generated for radar operations, and may be generated in accordance with radar operations performed by a radar module. For example, the audio signal may be generated by a 60 GHz Doppler radar module.

[0060] Method **1000** may perform operation **1006** during which the audio signal may be transmitted via one or more speakers within the automobile environment. Accordingly, the audio signal may be transmitted via one or more speakers included in the radar module, and may be transmitted to various locations within the automobile environment.

[0061] Method **1000** may perform operation **1008** during which an audio signal may be received via the microphone array and based, at least in part, on the plurality of phase parameters. Accordingly, a reflected audio signal may be received at the microphones included in the microphone array. The audio signal generated by the microphones may be phase modulated based on the phase modulation parameters to provide beam forming and directionality for listening capabilities of the microphone array. In this way a signal-to-noise ratio of received audio signals used for radar operations may be improved.

[0062] Method **1000** may perform operation **1010** during which one or more environmental parameters may be determined based on the received audio signal. As similarly discussed above, the environmental parameters may be generated based on radar computations, and may identify whether or not an entity is present at the location that was measured. More specifically, the environmental parameters may be configured to represent a positive or negative determination of a presence of an entity based on radar operations, and may be stored in memory.

[0063] Method **1000** may perform operation **1012** during which one or more operations may be performed based on the one or more environmental parameters. As similarly discussed above, the operations may be performed responsive to the detection of an entity. For example, in response to detecting an entity, such as a person, at the location being measured, a message may be generated and transmitted wirelessly to a wireless device, such as a smart phone. Accordingly, in response to detecting a person, operations such as generation of a notification message, lowering of automobile windows, and/or unlocking of doors may be performed.

[0064] FIG. **11** illustrates diagram of a method for automobile environment monitoring and signal steering, configured in accordance with some embodiments. As similarly discussed above, speakers may be configured to generate directional audio signals configured to target specific locations within an automobile environment. Moreover, microphones may be configured to directionally receive audio signals from locations. Accordingly, a method, such as method **1100**, may be performed to generate directional audio signals and directionally receive reflected audio signals used to improve radar operations within an automobile environment. In this way, beam forming and directionality provided by techniques disclosed herein may be leverage for both transmission and reception operations in radar determinations.

[0065] Method **1100** may perform operation **1102** during which a plurality of phase parameters may be determined for at least one speaker array and at least one microphone array. As similarly discussed above, such phase parameters may be determined based on a look-up-table configured for the at least one speaker array and the at least one microphone array, and the phase parameters may be determined by an entity, such as a manufacturer, during a design and manufacturing process. As also discussed above, an automobile environment may include multiple speaker arrays, such as four speaker arrays, and each speaker array may have a look-up-table configured specifically for that speaker array based on its position relative to locations within the automobile environment, such as seat locations.

[0066] Accordingly, during operation **1102**, a location may be identified, and phase parameters may be looked up based on the initial location that was identified. As similarly discussed above, a set of designated locations may be mapped to appropriate phase parameters within the look-up-table, and the phase parameters may be identified based on the mapping. Accordingly, the location may be provided to look-up-tables for the speaker arrays as well as a look-up-table for the microphone array. As also discussed above, the location may be identified based on an input received from the automobile environment. For example, such an input may be received from a sensor, such as a seatbelt sensor or a pressure sensor, included at a particular location, such as a seat location. In another example, the location may be identified based on a progression through a designated set of locations.

[0067] Method **1100** may perform operation **1104** during which an audio signal may be generated based, at least in part, on the plurality of phase parameters. As similarly discussed above, a waveform may be generated for radar operations, and may be modulated based on the phase parameters. For example, each speaker within the speaker array may have its own audio signal configured based on a phase parameter specific to that speaker. More specifically, the phase parameter may identify a phase shift or phase delay, and the audio signal for each speaker within the speaker array may be modified to implement the phase parameter for that speaker as determined based on the look-up-table.

[0068] Method **1100** may perform operation **1106** during which the audio signal may be transmitted via the speaker array within an automobile environment. Accordingly, the audio signal may be amplified and output via the speakers included in the speaker array. As discussed above, the phase parameters applied to the audio signals may provide beam forming and directionality of the audio output that targets a designated location that corresponds to the initially identified location mentioned above with reference to operation **1102**.

[0069] Method **1100** may perform operation **1108** during which an audio signal may be received via the microphone array and based, at least in part, on the plurality of phase parameters. Accordingly, a reflected audio signal may be received at the microphones included in the microphone array. The audio signals generated by the microphones may be phase modulated based on the phase modulation parameters to provide beam forming and directionality for listening capabilities of the microphone array. In this way a signal-to-noise ratio of received audio signals used for radar operations may be improved.

[0070] Method **1100** may perform operation **1110** during which one or more environmental parameters may be determined based on the received audio signal. As similarly discussed above, the environmental parameters may be generated based on radar computations, and may identify whether or not an entity is present at the location that was measured. More specifically, the environmental parameters may be configured to represent a positive or negative determination of a presence of an entity based on radar operations, and may be stored in memory.

[0071] Method **1100** may perform operation **1112** during which one or more operations may be performed. based on the one or more environmental parameters. As similarly discussed above, the operations may be performed responsive to the detection of an entity. For example, in response to detecting an entity, such as a person, at the location being measured, a message may be generated and transmitted wirelessly to a wireless device, such as a smart phone. Accordingly, in response to detecting a person, operations such as generation of a notification message, lowering of automobile windows, and/or unlocking of doors may be performed.

[0072] Although the foregoing concepts have been described in some detail for purposes of clarity of understanding, it will be apparent that certain changes and modifications may be practiced within the scope of the appended claims. It should be noted that there are many alternative ways of implementing the processes, systems, and devices. Accordingly, the present examples are to be considered as illustrative and not restrictive.

Claims

1. A method comprising: determining, using a processing device, a first plurality of phase modulation parameters for a speaker array based on a relationship between speakers included in the speaker array and a designated target location; generating, using the processing device, a plurality of audio signals for the speakers included in the speaker array, each of the plurality of audio signals including a phase adjustment determined based on the first plurality of phase modulation parameters; and transmitting the plurality of audio signals via the speaker array to the designated target location.

2. The method of claim 1, wherein the designated target location is identified based on one or more radar computations.

3. The method of claim 1, wherein the first plurality of phase modulation parameters is determined based on a look-up-table.

4. The method of claim 3, wherein the look-up-table provides a mapping between a plurality of designated target locations and a phase modulation parameter for each speaker for each of the plurality of designated target locations.

5. The method of claim 4, wherein the plurality of phase modulation parameters identify a phase offset value for each speaker included in the speaker array.

6. The method of claim 1 further comprising: determining, using the processing device, a second plurality of phase modulation parameters for a microphone array based on a relationship between microphones included in the microphone array and the designated target location; and receiving an audio signal via the microphone array from the designated target location.

7. The method of claim 6 further comprising: determining if an entity is present at the designated target location based, at least in part, on the received audio signal.

8. The method of claim 7, wherein the determining if the entity is present comprises: determining a temporal offset value associated with the received audio signal; and determining if an entity is present based on a comparison of the temporal offset value and a designated threshold value.

9. The method of claim 1, wherein the speaker array is included in an automobile.

10. A system comprising: a speaker array comprising a plurality of speakers; and a processing device comprising one or more processors configured to: determine a first plurality of phase modulation parameters for the speaker array based on a relationship between the plurality of speakers and a designated target location; generate a plurality of audio signals for the speakers included in the speaker array, each of the plurality of audio signals including a phase adjustment determined based on the first plurality of phase modulation parameters; and transmit the plurality of audio signals via the speaker array to the designated target location.

11. The system of claim 10, wherein the designated target location is identified based on one or more radar computations.

12. The system of claim 10, wherein the first plurality of phase modulation parameters is determined based on a look-up-table.

13. The system of claim 12, wherein the look-up-table provides a mapping between a plurality of designated target locations and a phase modulation parameter for each speaker for each of the plurality of designated target locations, and wherein the plurality of phase modulation parameters identify a phase offset value for each speaker included in the speaker array.

14. The system of claim 10, wherein the processing device is further configured to: determine a second plurality of phase modulation parameters for a microphone array based on a relationship between microphones included in the microphone array and the designated target location; and receive an audio signal via the microphone array from the designated target location.

15. The system of claim 14, wherein the processing device is further configured to: determine if an entity is present at the designated target location based, at least in part, on the received audio signal.

16. A device comprising: one or more processors configured to: determine a first plurality of phase modulation parameters for a speaker array based on a relationship between a plurality of speakers and a designated target location; generate a plurality of audio signals for the speakers included in the speaker array, each of the plurality of audio signals including a phase adjustment determined based on the first plurality of phase modulation parameters; and transmit the plurality of audio signals via the speaker array to the designated target location.

17. The device of claim 16, wherein the first plurality of phase modulation parameters is determined based on a look-up-table.

18. The device of claim 17, wherein the look-up-table provides a mapping between a plurality of designated target locations and a phase modulation parameter for each speaker for each of the plurality of designated target locations, and wherein the plurality of phase modulation parameters identify a phase offset value for each speaker included in the speaker array.

19. The device of claim 16, wherein the one or more processors are further configured to: determine a second plurality of phase modulation parameters for a microphone array based on a relationship between microphones included in the microphone array and the designated target location; and receive an audio signal via the microphone array from the designated target location.

20. The device of claim 19, wherein the one or more processors are further configured to: determine if an entity is present at the designated target location based, at least in part, on the received audio signal.